The design of parallel programs

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Credits

- prof. Salvatore Orlando, Univ. Ca' Foscari di Venezia
  http://www.dsi.unive.it/~orlando/
- prof. Mary Hall, University of Utah
  https://www.cs.utah.edu/~mhall/
- Tim Mattson, Intel

Sum-Reduction

- We start assuming a shared-memory architecture
  - All execution units share a common memory space
- We begin with a sequential solution and parallelize it
  - This is not always a good idea; some parallel algorithms have nothing in common with their sequential counterparts!
  - However, it is sometimes a reasonable starting point

Credits: Mary Hall

Sequential algorithm

- Compute the sum of the content of an array *A* of length *n*

```
float seq_sum(const float* v, int n)
{
    int i;
    float sum = 0.0;
    for (i=0; i<n; i++) {
        sum += v[i];
    }
    return sum;
}
```


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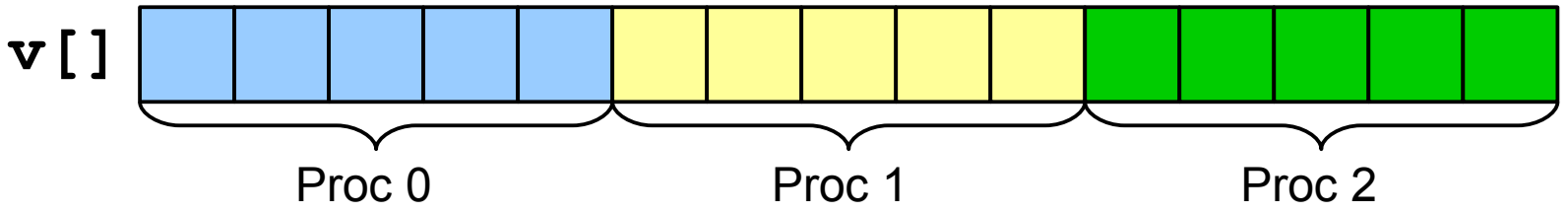
Version 1 Shared memory (Wrong!)

- Assuming *P* execution units (e.g., processors), each one computes a partial sum of *n / P* adjacent elements
- Example: *n* = 15, *P* = 3



WRONG

```
sum = 0.0;
do in parallel {
    my_block_len = n/P;
    my_start = my_id * my_block_len;
    my_end = my_start + my_block_len;
    for (my_i=my_start; my_i<my_end; my_i++) {
         sum += v[my_i];
    }
}
```

*Variables whose names start with my_ are assumed to be **local** (private) to each processor; all other variables are assumed to be **global** (shared)*

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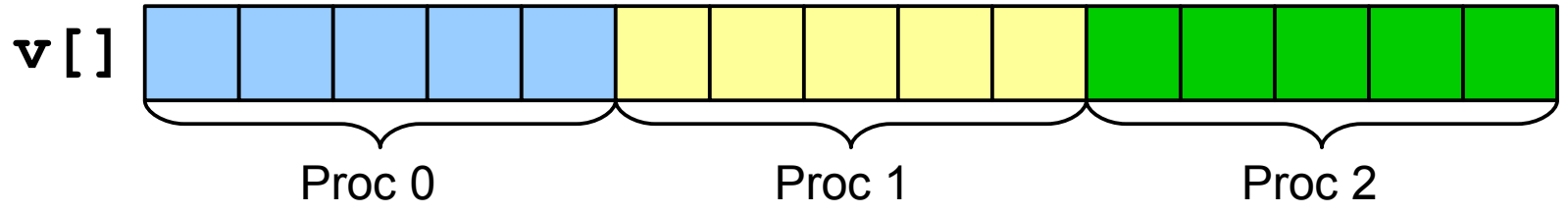
Version 1 (better, but still wrong)

- Assuming *P* processors, each one computes a partial sum of *n / P* adjacent elements
- Example: *n* = 15, *P* = 3

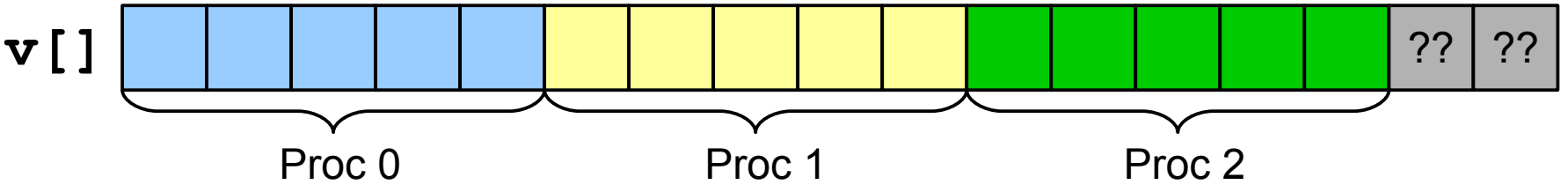


WRONG

```
mutex m;
sum = 0.0;
do in parallel {
    my_block_len = n/P;
    my_start = my_id * my_block_len;
    my_end = my_start + my_block_len;
    for (my_i=my_start; my_i<my_end; my_i++) {
        mutex_lock(&m);
        sum += v[my_i];
        mutex_unlock(&m);
    }
}
```

Version 1 (better, but still wrong)

- Assuming *P* processors, each one computes a partial sum of *n* / *P* adjacent elements
- Example: *n* = 17, *P* = 3

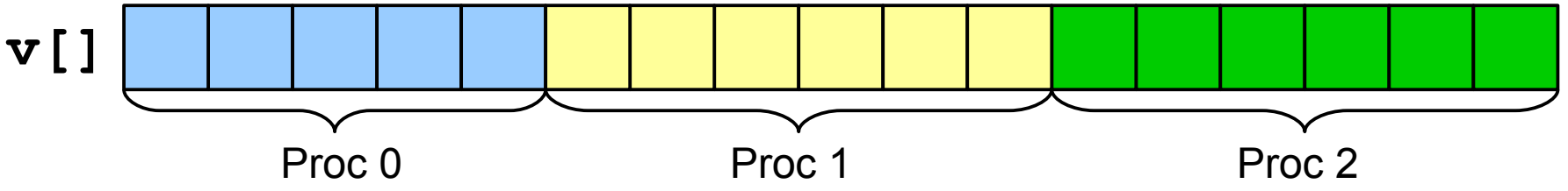


WRONG

```
mutex m;
sum = 0.0;
do in parallel {
    my_block_len = n/P;
    my_start = my_id * my_block_len;
    my_end = my_start + my_block_len;
    for (my_i=my_start; my_i<my_end; my_i++) {
        mutex_lock(&m);
        sum += v[my_i];
        mutex_unlock(&m);
    }
}
```

Version 1 (correct, not efficient)

- Assuming *P* processors, each one computes a partial sum of *n / P* adjacent elements
- Example: *n* = 17, *P* = 3




```
mutex m;
sum = 0.0;
do in parallel {
    my_start = (n * my_id) / P;
    my_end = (n * (my_id + 1)) / P;
    for (my_i=my_start; my_i<my_end; my_i++) {
        mutex_lock(&m);
        sum += v[my_i];
        mutex_unlock(&m);
    }
}
```

Version 2

- Too much contention on mutex *m*
- Solution: increase the mutex granularity
  - Each processor accumulates the partial sum on a local (private) variable
  - The mutex is used at the end to update the global sum

```
mutex m; sum = 0.0;
do in parallel {
    my_start = (n * my_id) / P;
    my_end = (n * (my_id + 1)) / P;
    my_sum = 0.0;
    for (my_i=my_start; my_i<my_end; my_i++) {
        my_sum += v[my_i];
    }
    mutex_lock(&m);
    sum += my_sum;
    mutex_unlock(&m);
}
```

Version 3: Remove the mutex
(wrong, in a subtle way)

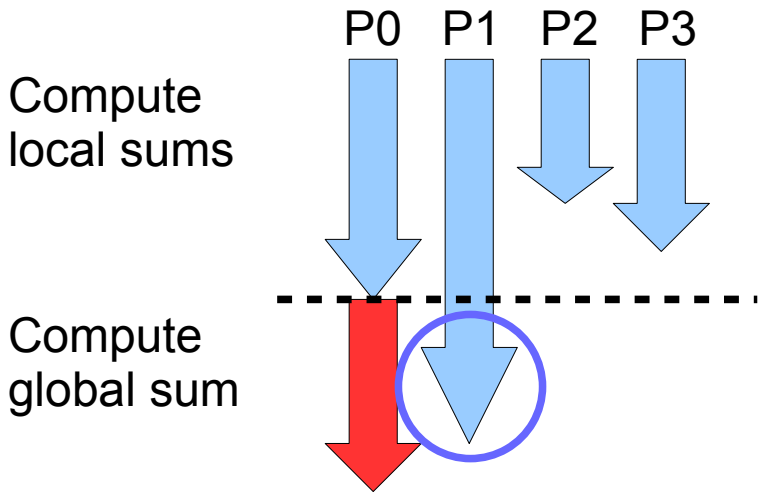
- We use a shared array **`psum[]`** where each processor can store its local sum
- At the end, one processor computes the global sum

WRONG

```
psum[0..P-1] = 0.0; /* all elements set to 0.0 */
sum = 0.0;
do in parallel {
    my_start = (n * my_id) / P;
    my_end = (n * (my_id + 1)) / P;
    for (my_i=my_start; my_i<my_end; my_i++) {
        psum[my_id] += v[my_i];
    }
    if ( 0 == my_id ) { /* only the master executes this */
        for (my_i=0; my_i<P; my_i++)
            sum += psum[my_i];
    }
}
```

The problem with version 3

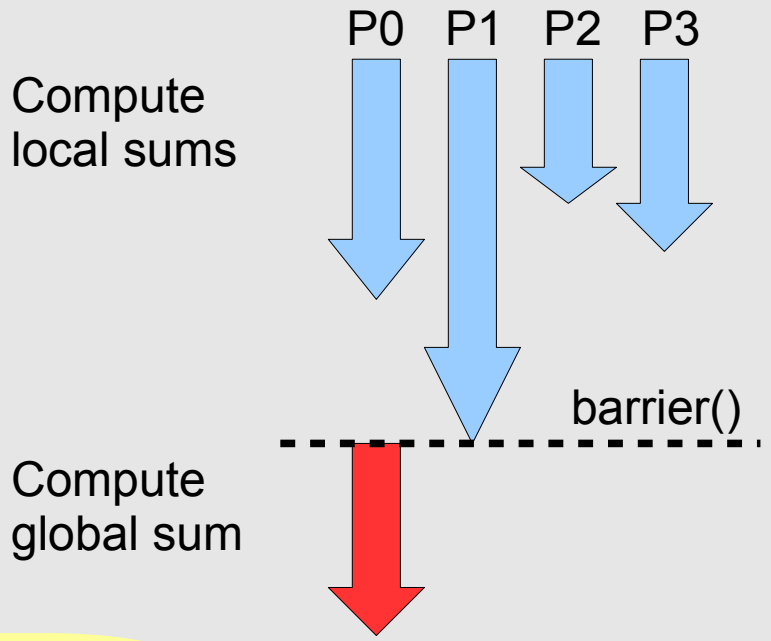
- Processor 0 could start the computation of the global sum before all other processors have computed the local sums!



Version 4 (correct)

- Use a barrier synchronization

```
psum[0..P-1] = 0.0;
sum = 0.0;
do in parallel {
    my_start = (n * my_id) / P;
    my_end = (n * (my_id + 1)) / P;
    psum[0..P-1] = 0.0;
    for ( my_i=my_start;
          my_i<my_end; my_i++  ) {
        psum[my_id] += v[my_i];
    }
    barrier();
    for (my_i=0; my_i<P; my_i++)
        sum += psum[my_i];
    }
}
```



Note: in OpenMP there is an implicit barrier at the end of a parallel region, so this pseudocode will be greatly simplified

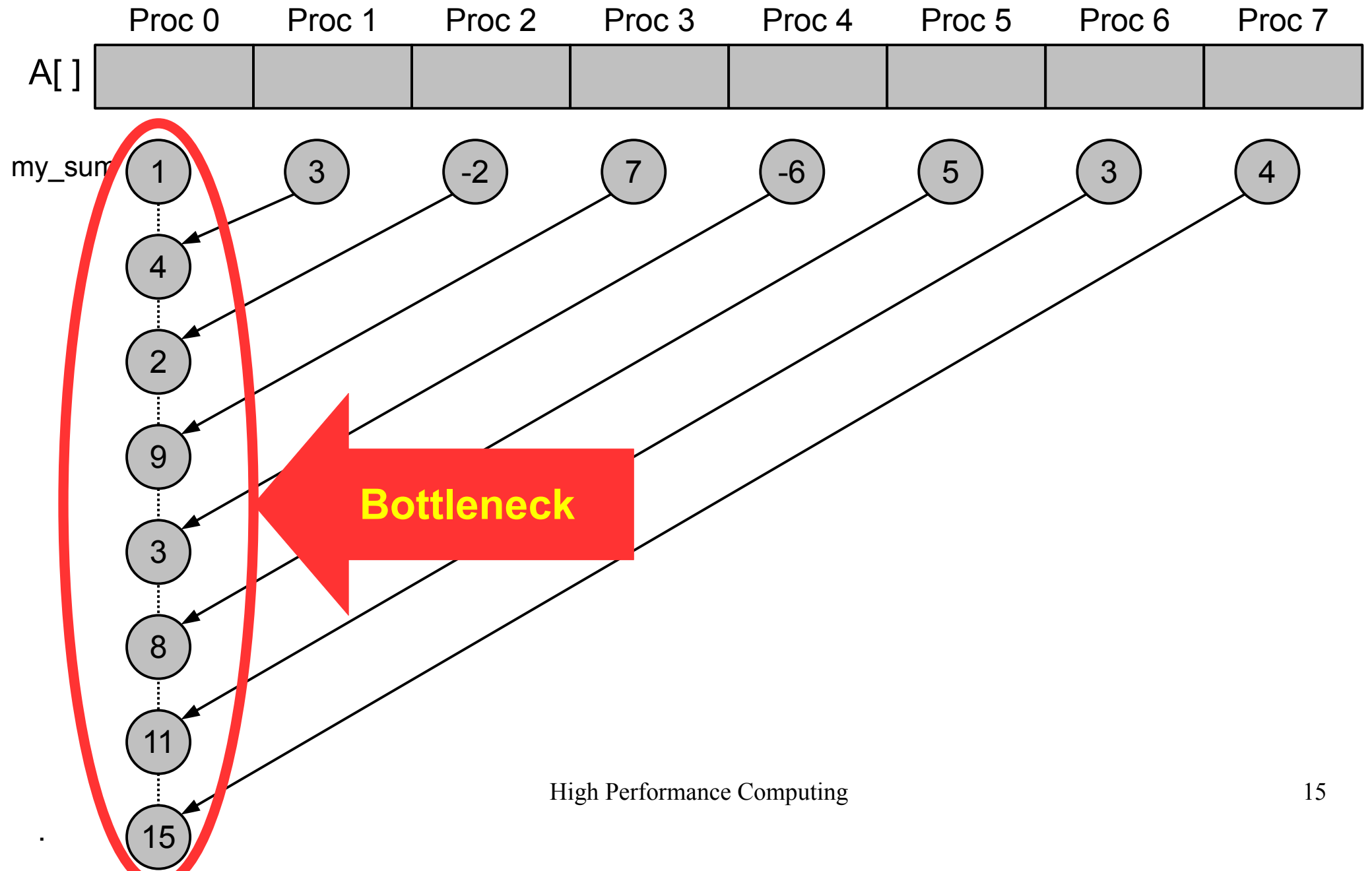
Version 5
Distributed-memory

- *P* << *n* processors
- All variables are private
- Each processor computes a local sum
- Each processor sends the local sum to processor 0 (the master)

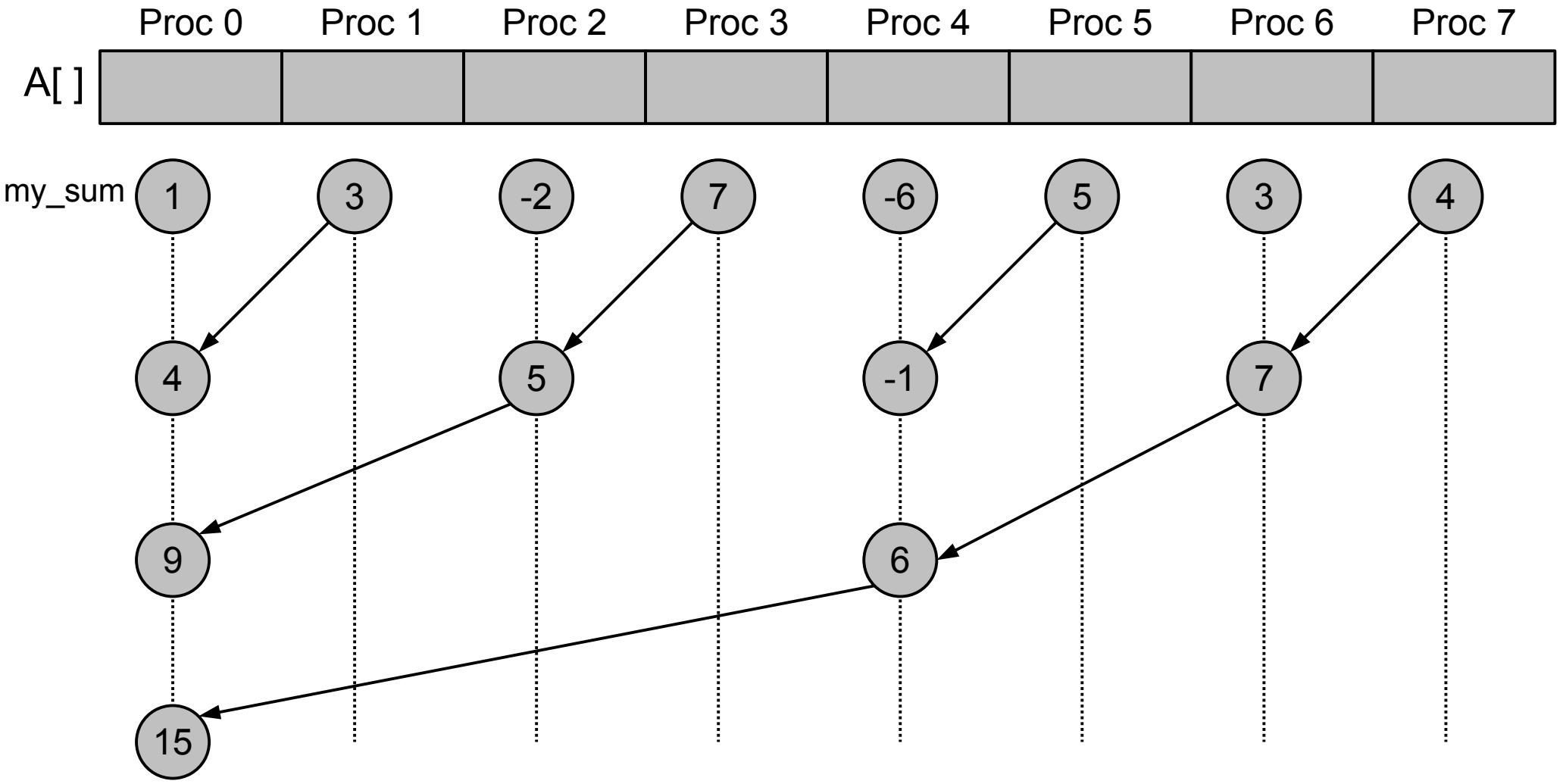
Executed by process my_id:

```
...
my_sum = 0.0;
my_start = …, my_end = …;
my_v[] = receive v[my_start..my_end-1]
from proc 0;
for ( i=0; i < my_end-my_start; i++ ) {
    my_sum += my_v[i];
}
if ( 0 == my_id ) {
    for ( i=1; i<P; i++ ) {
        tmp = receive from proc i;
        my_sum += tmp;
    }
    printf("The sum is %f\n", my_sum);
} else {
    send my_sum to proc 0;
}
```

Version 5


Proc 0
Proc 1
Proc 2
Proc 3
Proc 4
Proc 5
Proc 6
Proc 7
A[ ]
my_sum
1
3
-2
7
-6
5
3
4
4
2
9
3
8
11
15
Bottleneck

Parallel reduction



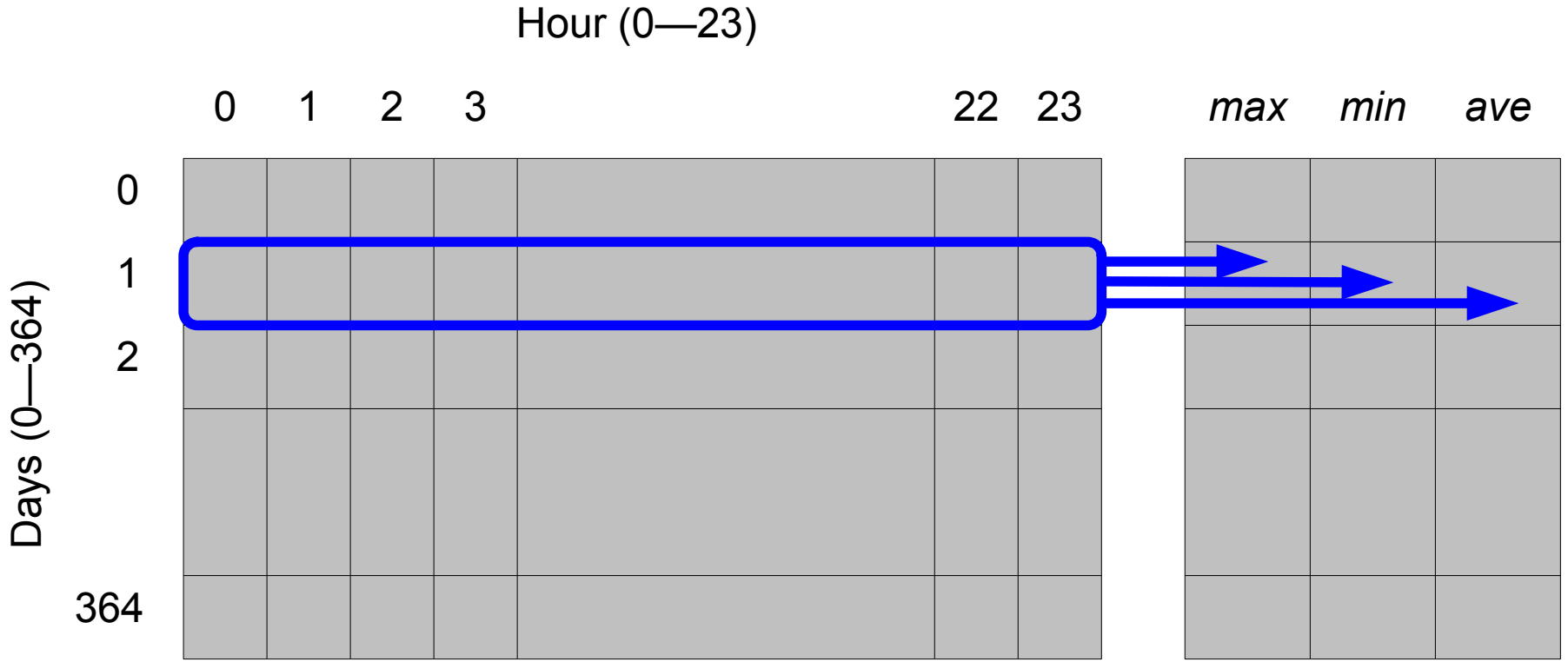
- (P – 1) sums are still performed; however, processor 0 receives ~ $\log_2 P$ messages and performs ~ $\log_2 P$ sums

Task parallelism vs Data parallelism

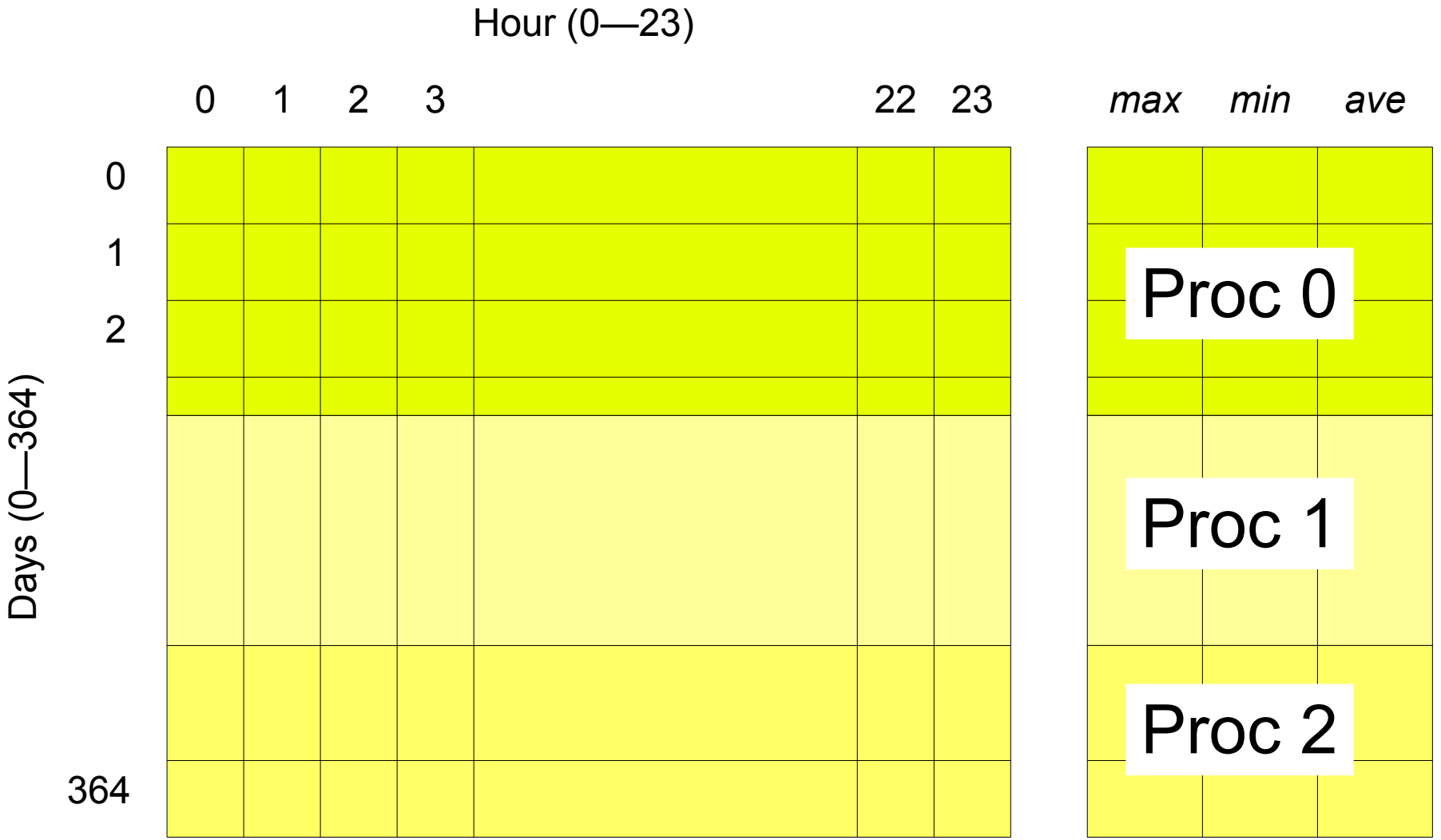
- Task Parallelism
  - Distribute (possibly different) tasks to processors
- Data Parallelism
  - Distribute data to processors
  - Each processor executes the same task on different data

Example

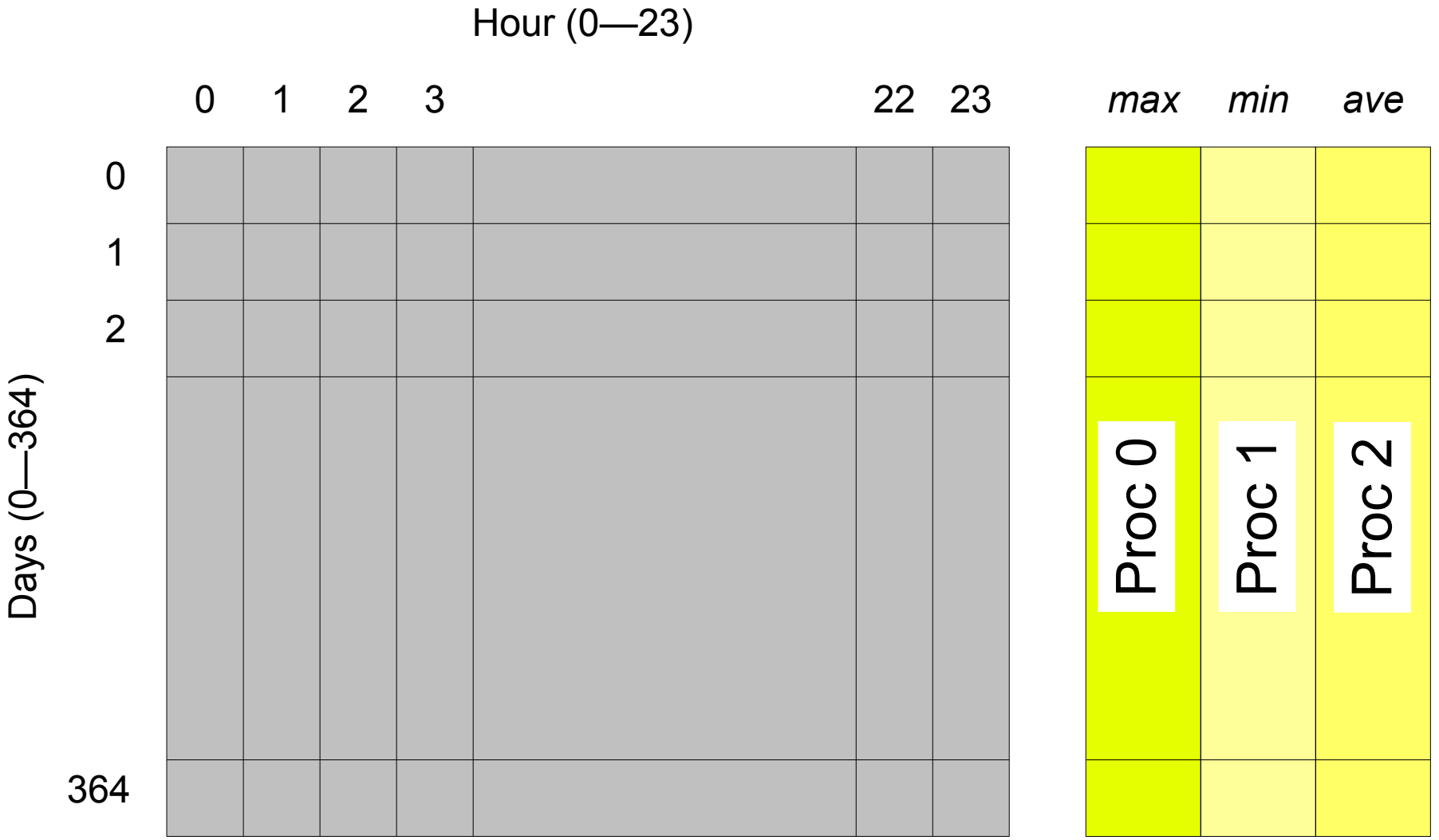
- Table containing hourly temperatures on some location
  - 24 columns, 365 rows
- Compute daily minimum, maximum and average
- Assume 3 execution units

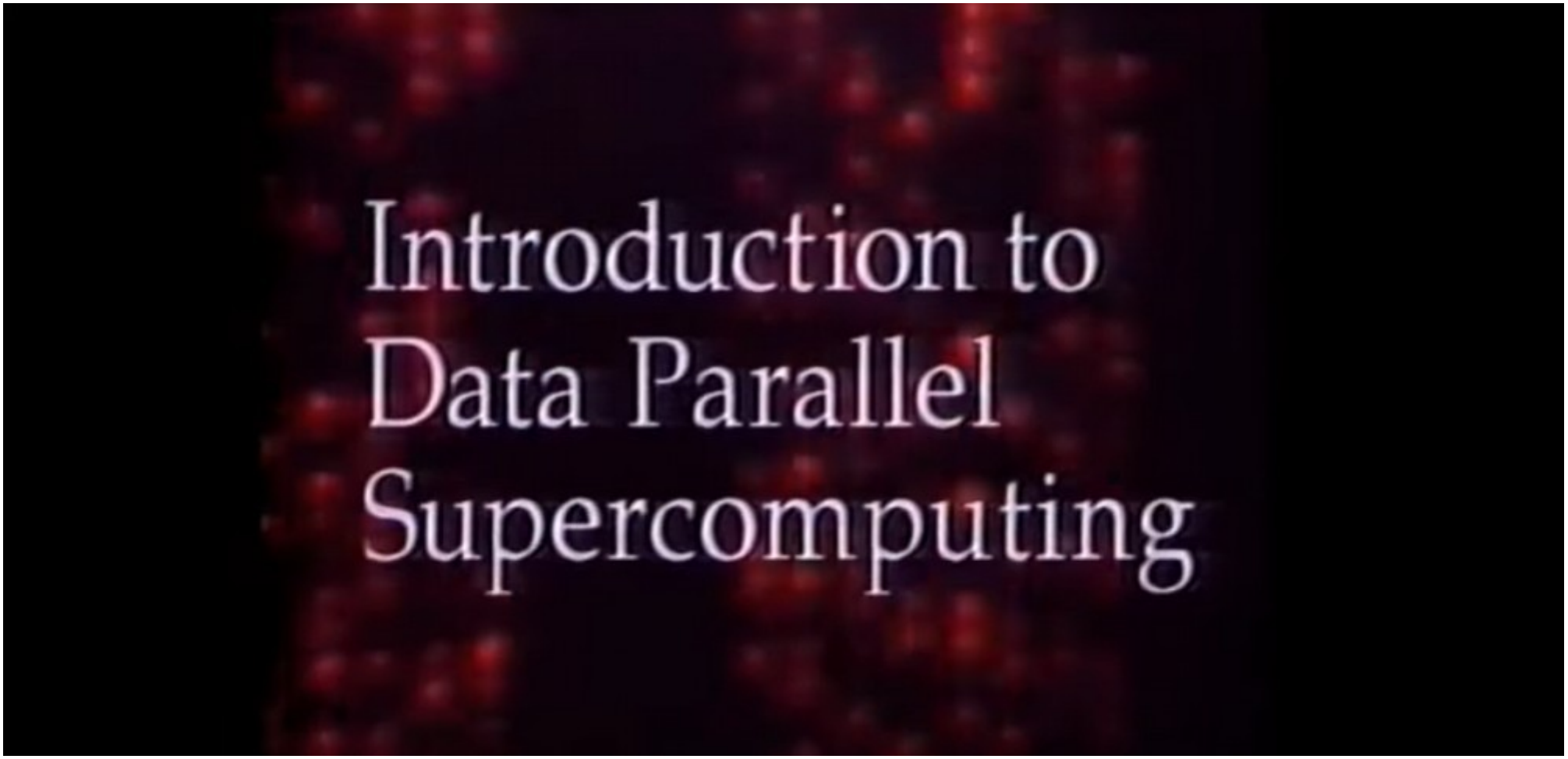


Data parallel approach



Task parallel approach





Thinking Machines Corporation (1989) , Introduction to Data Parallel Supercomputing, https://www.youtube.com/watch?v=f8NTHfMw_WA

Key concepts

- Parallel architectures require parallel programming paradigms
- Writing parallel programs is harder than writing sequential programs